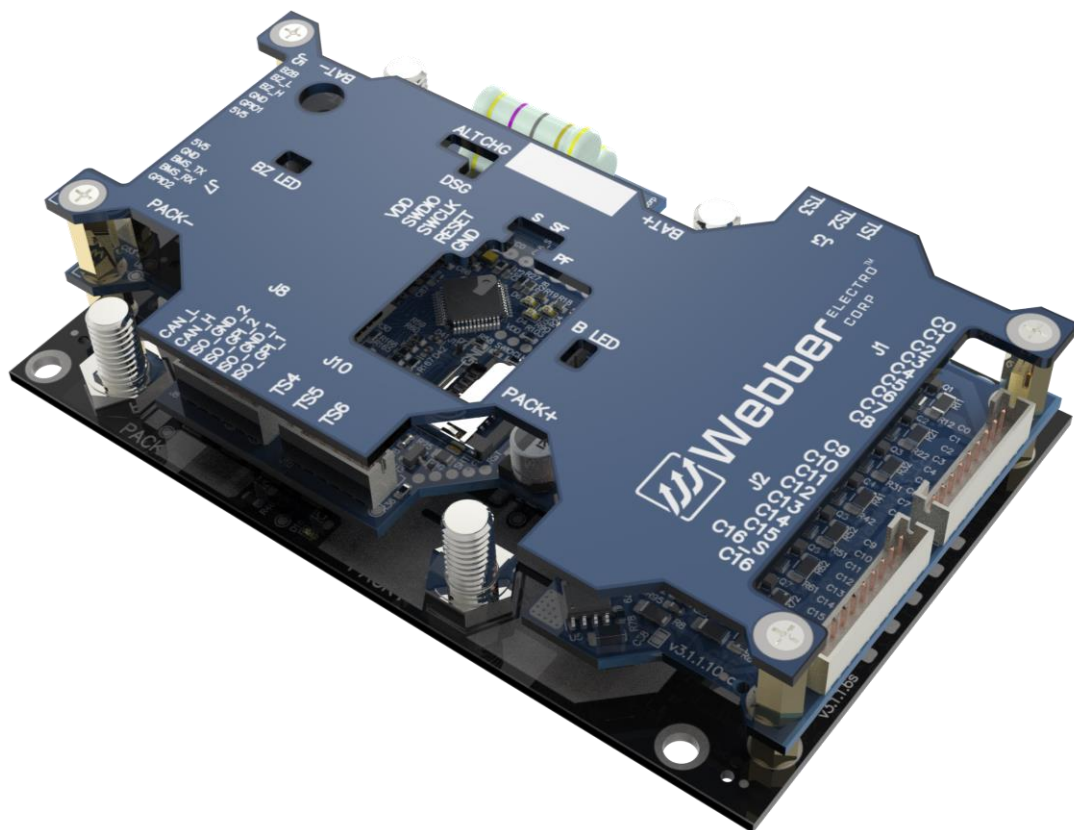


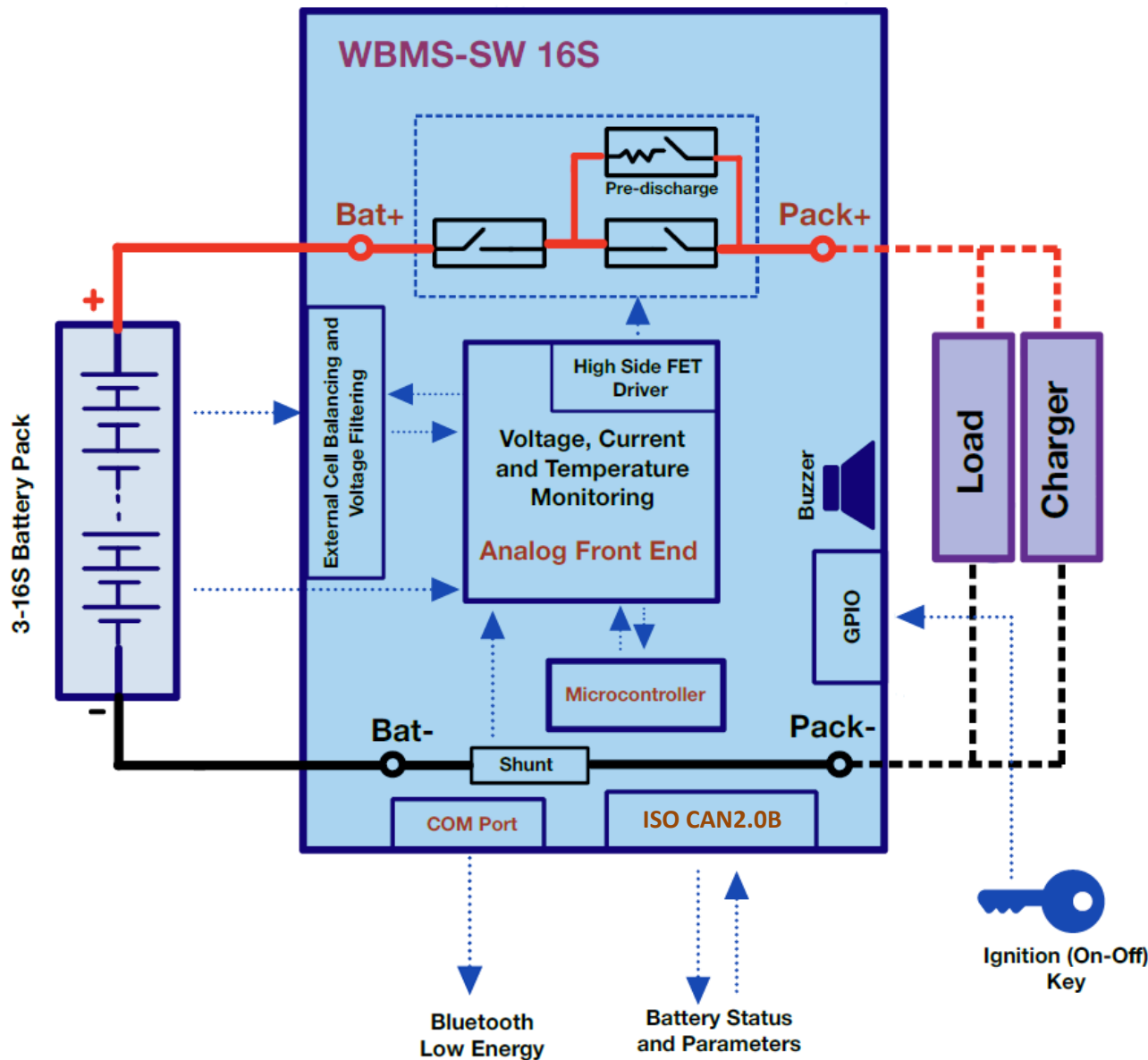
WBMS-SW-Mini 16S 45A

Smart BMS Manages Lithium Ion Batteries
with Precision and Speed

Specification Sheet v3.0



Application Diagram



Salient Features

- Scalable up to **16 cells** & **48-60V** battery packs
- **AIS-156 PH-II Compliant**
- **Buzzer output** and **LED Indication** for **Thermal Runaway, Open Wire Fault** and **MOSFET Failure**
- Multiple communication options: **Isolated CAN2.0** and **BLE**
- Onboard: **uSD Card** and **EEPROM**
- **High side N-MOSFET control** provides safe and uninterrupted ground connection to external subsystems.
- **Predischage control** for preventing inrush currents to capacitive loads. Prevents MOSFET failures and increases cell life
- **Short Circuit and Overcurrent in discharge latches** ensure automatic locking of discharge path after repeated short circuit and overcurrent faults
- **Cell balancing current up to 120mA** minimizes cell to cell voltage and SOC variations and optimal charging of battery pack
- Individual Cell over and under voltage and temperature protection
- Pack level over current and short circuit protection
- **Deep Discharge Prevention (Auto BMS Shutdown)** when cell voltage is critically low
- Proprietary **CAN based field servicing tools** for **data monitoring, logging** and **firmware update**

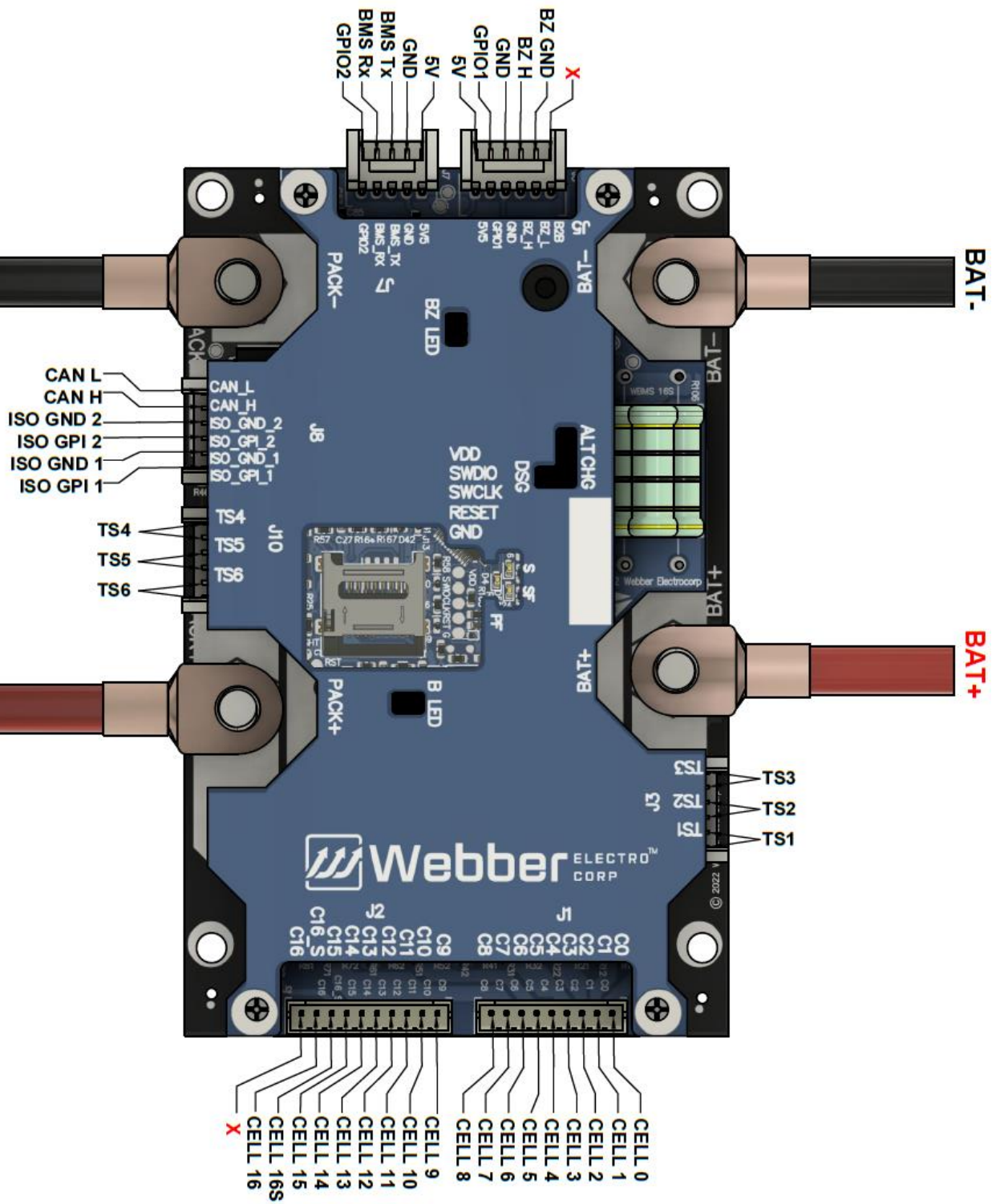
Specifications		Remarks	
Number of Cells		Min: 3 Max: 16	
Cell Voltage Sensing Range		1.6 - 4.5V	
Peak Battery Terminal Voltage		68 V (16S)	
Sensing Accuracy		Cell Voltage: $\pm 3\text{mV}$ Current: $\pm 1\%$	
Dimension		115 mm X 72 mm X 21 mm (L X B X H)	
Rated Nominal Continuous Current Rating	Charging and Discharging Current	Up to 45A . Needs thermal connection to metal battery pack casing for PDU heat dissipation to maintain continuous current rating. Continuous current rating to be derated to ~65% of the rated value if no thermal connection is present.	
Cell Balancing	Balancing Current per Cell	WBMS-SW-Mini-LB: 50~60 mA WBMS-SW-Mini-HB: 100~120 mA	
Pre-discharge control	Nominal Capacitive load	3-6mF in 0.5 seconds	
BMS Self Current Consumption	Active Mode, Shutdown Mode	Active Mode: ~13mA Shutdown Mode: ~0.15mA	
Temperature Sensing	Onboard	One channel for onboard MOSFET temperature	
	External	Six external channels for cell temperature	
	Range and Accuracy	Range: -10°C to 120°C Accuracy $\pm 1^\circ\text{C}$,	
Onboard Data Storage	uSD Card or EEPROM	uSD Card: every 1 second, logged over last 60 days EEPROM: every 90 seconds, logged over last 30 days	
Over Voltage Protection	Threshold, Protection Trigger Delay	Configurable	
	Fault Response	Charge MOSFET: OFF, Discharge MOSFET: ON	
	Fault Release	Through Hysteresis, Automatic	
Under Voltage Protection	Threshold, Protection Trigger Delay	Configurable	
	Fault Response	Discharge MOSFET: OFF, Charge MOSFET: ON	
	Fault Release	Configurable Hysteresis	No load removal required. Charging current is allowed.
Over Current in Discharge Protection Level-1	Threshold, Protection Trigger Delay	Configurable to allow up to 80A discharge for up to 60 seconds	
	Fault Response	Discharge MOSFET: OFF, Charge MOSFET: ON	
	Fault Release	Auto recovery after no discharge current for 90 secs	
Over Current in Discharge Protection Level-2	Threshold, Protection Trigger Delay	Configurable to instantaneously cut off at 120A discharge current in 100 mSec	
	Fault Response	Discharge MOSFET: OFF, Charge MOSFET: ON	
	Fault Release	Auto recovery after no discharge current for 90 secs	

Note: Specifications are indicative only. Final specifications and behaviour including continuous discharge/charge current, balancing current, protection thresholds, etc. may vary depending upon BMS version, BMS mounting scheme, PDU heat dissipation, programmed parameters and application logic.

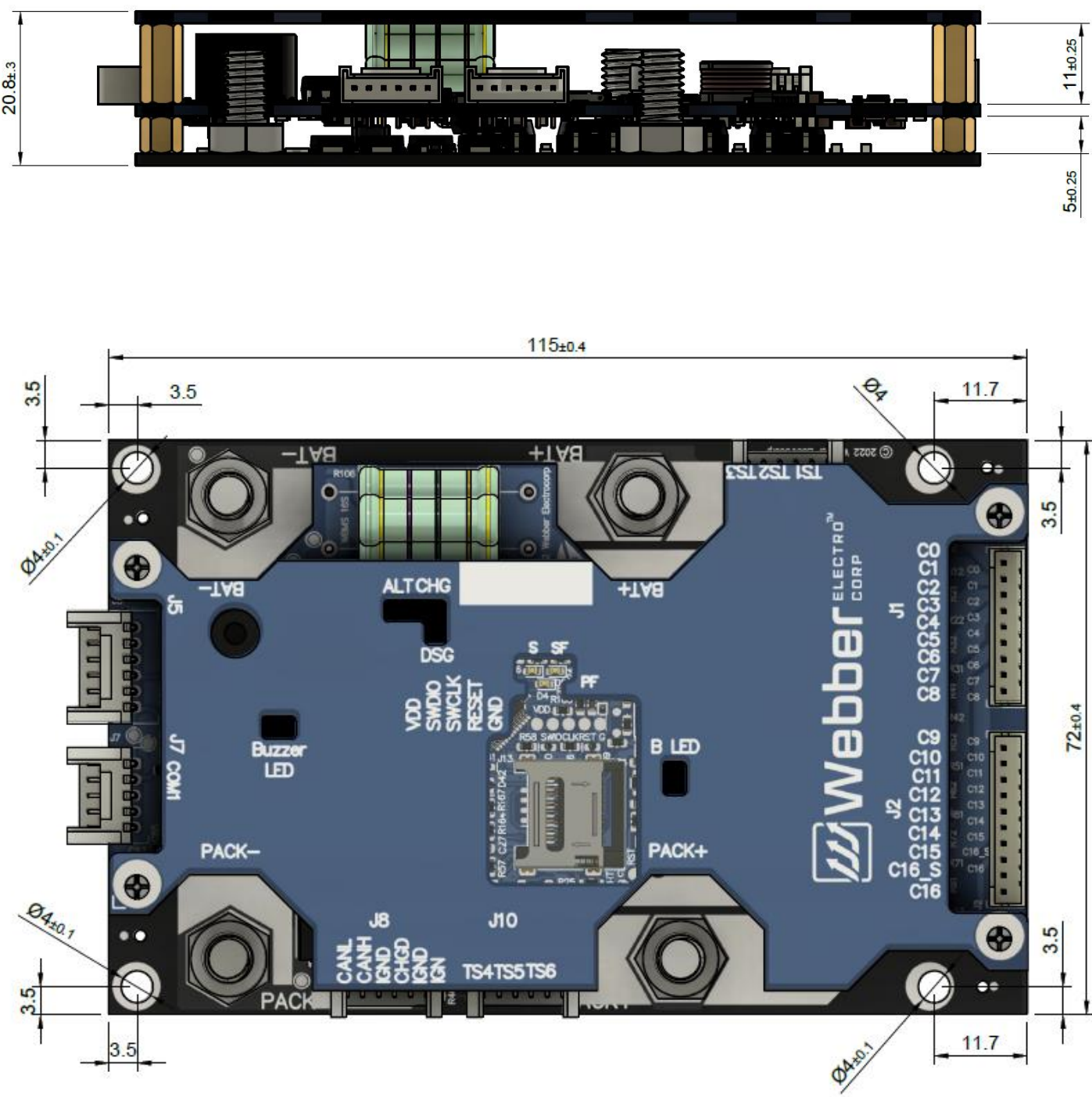
Specifications			Remarks
Over Current in Discharge Latch Protection	Threshold	5 consecutive over discharge current faults	
	Fault Response	Discharge MOSFET: OFF, Charge MOSFET: ON	
	Fault Release	Load Removal or Auto release	
Over Current in Charge Protection Level-1	Threshold, Protection Trigger Delay	Configurable to allow up to 45A charge current for up to 60 seconds	
	Fault Response	Discharge MOSFET: ON, Charge MOSFET: OFF	
	Fault Release	Auto recovery after no charge current for 90 secs	
Over Current in Charge Protection Level-2	Threshold, Protection Trigger Delay	Configurable to instantaneously cut off at 64A charge current in 100 mSec	
	Fault Response	Charge MOSFET: OFF, Discharge MOSFET: ON	
	Fault Release	Auto recovery after no charge current for 90 secs	
Short Circuit in Discharge Protection	Protection Trigger Delay	~30 uSec	
	Fault Response	Discharge MOSFET: OFF, Charge MOSFET: OFF	
	Fault Release	Auto recovery after 5 seconds	
Short Circuit in Discharge Latch Protection	Threshold	2 consecutive short circuit faults	
	Fault Response	Discharge MOSFET: OFF, Charge MOSFET: OFF	
	Fault Release	Load Removal	
Over Temperature in Discharge Protection	Threshold, Protection Trigger Delay	Configurable	
	Fault Response	Discharge MOSFET: OFF, Charge MOSFET: ON	
	Fault Release	Auto recovery below configurable temperature	
Under Temperature in Discharge Protection	Threshold, Protection Trigger Delay	Configurable	
	Fault Response	Discharge MOSFET: OFF, Charge MOSFET: ON	
	Fault Release	Auto recovery above configurable temperature	
Over Temperature in Charge Protection	Threshold, Protection Trigger Delay	Configurable	
	Fault Response	Charge MOSFET: OFF, Discharge MOSFET: ON	
	Fault Release	Auto recovery below configurable temperature	
Under Temperature in Charge Protection	Threshold, Protection Trigger Delay	Configurable	
	Fault Response	Charge MOSFET: OFF, Discharge MOSFET: ON	
	Fault Release	Auto recovery above configurable temperature	
Buzzer Output	Alert on Permanent Failure	Thermal Runaway, MOSFET Failure, Cell Open Wire, Short Circuit Latch, Over Current in Discharge Latch, Insulation Failure	

Product Naming Convention WBMS-SW-Mini-AB-16S-45A-BBBB-CCCC-VD		
A Balancing Current	= L (WBMS-SW-Mini-LB-16S-45A-)	Balancing Current = 50~60mA
	= H (WBMS-SW-Mini-HB-16S-45A-)	Balancing Current = 100~120mA
BBBB CAN Communication Hardware Option	= NISO (WBMS-SW-Mini-AB-16S-45A-NISO-)	Non Isolated CAN
	= ISO (WBMS-SW-Mini-AB-16S-45A-ISO-)	Isolated CAN
CCCC Onboard Data Storage Option	= MSDH (WBMS-SW-Mini-AB-16S-45A-BBBB-MSDH)	uSD Card Holder only
	= MSDHC (WBMS-SW-Mini-AB-16S-45A-BBBB-MSDHC)	uSD Card Holder + uSD Card
	= EPRM (WBMS-SW-Mini-AB-16S-45A-BBBB-EPRM)	EEPROM
D Hardware Version	= null (WBMS-SW-Mini-AB-16S-45A-BBBB-CCCC)	Initial release with multiple daughter boards
	= 2.0 (WBMS-SW-Mini-AB-16S-45A-BBBB-CCCC-V2.0)	Latest release two board architecture with onboard buzzer

LOAD+



Mechanical Dimensions



All dimensions in mm